

Chutong Yang

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Research Focus

Learning theory and algorithm design, with an emphasis on trustworthy machine learning (calibration, algorithmic fairness, and differential privacy) and optimization.

Recent work focuses on uncertainty quantification and reliable LLM reasoning, including uncertainty-guided multi-agent deliberation, RL-based confidence calibration, and mathematical reasoning with LLMs; broader interests include autonomous agent harnesses.

Education

The University of Texas at Austin , Ph.D. in Computer Science • Recipient, Amazon AI PhD Fellowship	Aug 2023 – Present
Stanford University , M.S. in Computer Science • Full financial support for two years.	Sep 2021 – Jun 2023
University of California, San Diego , B.S. in Computer Science and Mathematics	Sep 2016 – Jun 2020

Experience

Amazon Web Services (AWS) , Applied Scientist Intern • Built a benchmark of sandboxed theoretical computer science (TCS) challenges, enabling controlled comparisons of models and agentic system designs. • Designed and evaluated multiple agent harnesses, including plan-and-execute and Monte Carlo tree search (MCTS) approaches.	Summer 2026
The University of Texas at Austin , Research Assistant • Conduct research with Prof. Kevin Tian on learning theory and algorithm design, focusing on calibration, algorithmic fairness, differential privacy, and optimization. • Develop methods for reliable LLM reasoning through uncertainty-guided multi-agent deliberation and RL-based confidence calibration.	Aug 2023 – Present
Stanford University , Research Assistant • Researched algorithmic fairness with Prof. Omer Reingold, focusing on multicalibration. • Developed optimization algorithms with Prof. Aaron Sidford using interior-point methods.	2022 – 2023

Skills

Programming: Python, C++ , Java, MATLAB
ML/RL: PyTorch, verl

Publications

(Authors are listed alphabetically unless otherwise indicated by *.)

11. Optimal Recalibration of an Online Predictor Lunjia Hu, Kevin Tian, <i>Chutong Yang</i> https://arxiv.org/abs/2607.19689	Preprint, 2026
10. Simultaneous Blackwell Approachability and Applications to Multiclass Omniprediction Lunjia Hu, Kevin Tian, <i>Chutong Yang</i> https://arxiv.org/abs/2602.17577	COLT 2026

- 9. Spherical Leech Quantization for Visual Tokenization and Generation** CVPR 2026 (Highlight)
Yue Zhao*, Hanwen Jiang, Zhenlin Xu, *Chutong Yang*, Ehsan Adeli, Philipp Krähenbühl
<https://arxiv.org/abs/2512.14697>
- 8. Proportionality from Low-Dimensional Approval Data** AAMAS 2026
Zhiyi Huang, Gregory Kehne, *Chutong Yang*
Extended abstract: <https://doi.org/10.65109/YHKI7695>
- 7. Private Geometric Median in Nearly-Linear Time** NeurIPS 2025
Syamantak Kumar, Daogao Liu, Kevin Tian, *Chutong Yang*
TPDP 2025 Workshop
<https://arxiv.org/abs/2505.20189>
- 6. Omnipredicting Single-Index Models with Multi-Index Models** STOC 2025
Lunjia Hu, Kevin Tian, *Chutong Yang*
<https://arxiv.org/abs/2411.13083>
- 5. Testing Calibration in Nearly-Linear Time** NeurIPS 2024
Lunjia Hu, Arun Jambulapati, Kevin Tian, *Chutong Yang*
<https://arxiv.org/abs/2402.13187>
- 4. Omnipredictors for Constrained Optimization** ICML 2023
Lunjia Hu, Inbal Livni-Navon, Omer Reingold, *Chutong Yang*
<https://arxiv.org/abs/2209.07463>
- 3. Active Learning Polynomial Threshold Functions** NeurIPS 2022
Omri Ben-Eliezer, Max Hopkins, *Chutong Yang*, Hantao Yu
<https://arxiv.org/abs/2201.09433>
- 2. A Fast Exact Algorithm for Deployment of Sensor Nodes for Internet of Things** ISF 2020
Qinghua Zheng*, *Chutong Yang*, Haijun Yang, Jianhe Zhou
Information Systems Frontiers: <https://link.springer.com/article/10.1007/s10796-018-9890-3>
- 1. Detailed placement for IR drop mitigation by power staple insertion in sub-10nm VLSI** DATE 2019
Sun ik Heo, Andrew B Kahng, Minsoo Kim, Lutong Wang, *Chutong Yang*
<https://ieeexplore.ieee.org/abstract/document/8715096>

Professional Service

Conference reviewer: NeurIPS 2024, ICLR 2025, ICML 2025, ITCS 2026